

Topics

This exam covers everything in the course. Here is a list of topics covered so far in the course. Last updated: **July 8, 2026**

1. Ordered Fields

- Examples/Non-examples
- Definition and properties of $\leq$

2. The Real Line

- Definition and properties of the absolute value
- Definition and properties of $\sup/\inf$
- The completeness axiom
- Archimedean property
- Dense sets

3. Metric Spaces

- Examples of metric spaces.
- Norms induce metrics
- Open/Closed sets in metric spaces
- Bounded sets.
- Interior, Closure, and Boundaries
- Equivalent norms and metrics

4. Sequences in Metric Spaces

- Bounded and convergent subsequences
- Subsequences

5. Sequences in $\mathbb{R}$

- Properties of convergent subsequences
- Monotone sequences
- Definition and properties of $\limsup/\liminf$

6. Completed Metric Spaces

- Cauchy sequences
- Definition of a complete metric space
- $\mathbb{R}$ is complete
- $\mathbb{R}^d$ is complete

7. Topological Spaces

- Definition of a Topological Space $(X, \mathcal{T})$.
- Metric Topology
- Convergence in Topological spaces
- Definition compact sets
- Closed $\cap$ compact is closed
- Nested sequences of compact sets have non-empty compact intersection.

8. Compact Sets in Metric Spaces.
 - Definition of Sequentially compact
 - Definition of Totally Bounded
 - examples and non-examples of seq
 - Compact $\implies$ Seq. Compact $\implies$ Totally Bounded and Complete $\implies$ Compact.
 - Heine-Borel Theorem.
9. Continuous functions
 - Definition and equivalent definitions
 - Cluster points
 - Images of compact sets are compact
 - Properties of real-values continuous functions (e.g. $f + g$ is continuous.)
10. Uniform continuity.
 - Definition
 - Examples/Non-examples
 - For uniformly continuous f , x_n is Cauchy $\implies f(x_n)$ is Cauchy.
 - Continuous functions with compact domain are uniformly continuous.
11. Sequences of functions and types of convergence
 - Definition of Point-wise convergence
 - Example where the limit of sequence of functions is not continuous.
 - Definition of uniform convergence
 - Example of uniformly converging sequence of functions.
 - Uniform Cauchy Criterion
 - For complete co-domains, Uniformly Cauchy $\iff$ Uniform convergence
12. Connected sets
 - Definitions
 - $[0, 1]$ is connected.
 - Definition of path connected.
 - Images of connected sets are connected.
 - Examples of path connected and connected sets.
 - Path connected implies connected. Converse is not true.
 - Characterization of connected sets in $\mathbb{R}$.
 - Open and connected in $\mathbb{R}^n$ implies path connected.
 - The topologist's curve. (Do NOT need to know **how** to show it is connected but not path connected)
13. Differentiation
 - Definition of the derivative
 - Differentiation rules
 - Mean Value Theorem and Rolle's Theorem
 - Corollaries of MVT and Rolle's
 - Inverse function theorem
 - Taylor's theorem and Corollaries

14. Integration

- Partitions
- Upper and lower Darboux sums
- Darboux integrable
- Cauchy Criterion for integrals
- Riemann integrable
- Continuous $\implies$ integrable
- The integral is linear
- The composition of a continuous function and integrable function is integrable.
- Fundamental theorems of calculus
- logarithm and exponentiation.
- change of variables

15. Interchanging limit operations.

- Uniform convergence and integrals
- Uniform convergence and differentiation.
- Leibniz rule(not examinable).

16. Series

- Definition
- Absolute convergence and conditional convergence
- Rearranging series
- Tests of convergence(Comparison, Root, Ratio, and **Integral**)

17. Power series

- Definition
- Examples from Taylor series
- Radius of convergence
- Weierstrass M-test
- Integral/Derivatives of powers series

Questions

1. Assume s_n and t_n are sequences of real numbers. If $s_n \rightarrow s > 0$ then

$$\limsup s_n t_n = s \limsup t_n.$$

Why does it matter that $s \neq 0$?

2. Let B be a topological space. Suppose $f : B \rightarrow R$ is continuous. Prove or disprove: If $f(B)$ is connected than B is connected.
3. Let X, Y be metric spaces. Suppose that $f : X \rightarrow Y$ is an injective continuous function. Prove that if $f(X)$ is connected then so is X .
4. Let X, Y , and Z be metric spaces and $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be maps. Assume further that:
- X is compact;

- f is surjective and continuous; and
- $g \circ f$ is continuous.

Show that g is continuous.

5. Determine whether the function

$$f(x) = |x|$$

is differentiable at $x = 0$.

6. Suppose f is differentiable on (a, b) and continuous on $[a, b]$. Prove that if

$$f'(x) > 1 \quad \text{for all } x \in (a, b),$$

then

$$f(b) - f(a) > b - a.$$

7. Let $f(x)$ be continuous at 0. Show that $xf(x)$ is differentiable at 0.

8. From first definition prove that x^2 is Darboux integrable.

9. Prove the Integral Test for series convergence.

10. Prove that the following series converge:

- (a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$
- (b) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$
- (c) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$
- (d) $\sum_{n=1}^{\infty} \frac{\sin(n)}{n^2}$

11. Compute the radius of convergence of the following power series:

- (a) $\sum_{n=0}^{\infty} \frac{x^n}{n!}$
- (b) $\sum_{n=1}^{\infty} \frac{3^n}{\sqrt{n}} x^n$
- (c) $\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2} x^n$
- (d) $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{n^2} x^n$

12. Let f be a bounded integrable function from $[a, b] \rightarrow \mathbb{R}$. Prove that

$$F(x) := \int_a^x f(t) \, dt$$

is continuous on $[a, b]$.

13. Consider the infinite series of functions defined on the interval $[0, \pi]$ by:

$$S(x) = \sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$$

- (a) Prove that the series converges uniformly on $[0, \pi]$ to a continuous function $S(x)$. (*Hint: Use the Weierstrass M-test.*)
- (b) Justify why the integral and summation operations can be interchanged, and evaluate the integrated series explicitly:

$$\int_0^\pi \sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2} \, dx = \sum_{n=1}^{\infty} \int_0^\pi \frac{\sin(nx)}{n^2} \, dx$$

14. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a uniformly continuous function, and let

$$f_n(x) = \frac{n}{2} \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} f(t) \, dt.$$

Show that $f_n \rightarrow f$ uniformly on $\mathbb{R}$.